A market such as Turkey or Vietnam, weakly correlated with the others, mechanically appears as a receiver. Trading hours add a second source of bias, since all funds trade in New York after the Asian markets have closed. The idiosyncratic network gives a different ranking (Spearman 0.30 with Diebold–Yilmaz): Taiwan and Indonesia are the main transmitters and Argentina is close to neutral. Argentina is thus exposed to the common emerging-market shock rather than to shocks from specific neighbors, and its largest country-specific shock, the August 2019 primary elections, does not turn it into a transmitter in any of the four systems: in our data, that local shock did not spread.

Limitations. First, we study returns, not the realized volatilities of the base paper, so our networks describe the transmission of return shocks, not of risk. Second, ETF prices embed exchange-rate movements against the dollar and US trading hours, which affect the estimated directions. Third, the rolling windows of 104 weekly observations are short for the sparse estimators, whose results depend on tuning choices: NETS is selected by BIC over a grid of 15 values, which makes the series jump from one window to the next, and FNETS requires a positive-definite adjustment of the idiosyncratic covariance, gives an empty network in the full sample, and shows one-window drops that coincide with changes in the number of retained coefficients. With $N = 18$, the cross-section is also small for the factor-number criterion and for the consistency of the dynamic principal components, which rely on large N. Fourth, almost no lagged links survive in the sparse methods, so the Granger networks carry little information. Fifth, consecutive windows share 100 of their 104 weeks, so the stability measures are high partly by construction; they are informative for comparing methods but not as absolute values. Finally, we report no confidence intervals, so small differences, such as 81.0% against 80.0% between Diebold–Yilmaz and the lasso VAR, should not be interpreted as significant.

7 Conclusion

Measured connectedness among emerging equity markets depends less on the estimation method than on whether the common factor is removed. At the weekly frequency connectedness is high, around 80% for the dense measures, and almost entirely contemporaneous; the lasso VAR and Diebold–Yilmaz agree closely, and NETS reproduces the same pattern at a lower level. A single common factor explains more than half of the variance of returns, and removing it lowers connectedness by about 40%, a share that is stable across windows, horizons and samples; the peak of 2020 and the decline afterwards reflect changes in the weight of this factor rather than in bilateral links. Most of the co-movement of emerging markets is therefore common exposure, not contagion. What cannot be claimed with the same confidence is the direction of spillovers between countries, which our data barely identify, and the structure of the idiosyncratic network, which is unstable in short windows. Natural extensions are to study volatility connectedness, as in the base paper, to use higher-frequency data or longer windows for the factor-adjusted estimators, and to add more markets, which would also improve the estimation of the common factor.

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9 Data and Code Availability

The data (weekly ETF returns) and the full code are available at `https://github.com/ezellabra/ML4E-code`. The pipeline is written in Python; NETS and FNETS are estimated in R (packages `nets` 0.9.1 and `fnets` 0.1.6) through a command-line bridge. All parameters are in `config.yaml` and the whole analysis is reproduced with `make all` (or `python run_all.py`).

A Robustness

Table 3 reruns every method under alternative choices. The levels of Diebold–Yilmaz and lasso connectedness hardly move: between 79.7% and 81.2% and between 78.6% and 80.5% across all scenarios. NETS is the most sensitive to the window length (57.8% with 78-week windows, 63.4%